



CAT 6 FLASHER 12 V LOW TEMPERATURE OPERATING TEST REPORT

DATE:22.05.2026

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Part Name	CAT 6 FLASHER 12 V	Application	COMMERCIAL APPLICATION
Customer Part No.	B00 861 544 00 32	Customer	FORCE MOTOR
UCAL Part No.	400230122002	Report No	R&D/CAT6/12VF/DUR/26/034

INTRODUCTION:

UCAL is currently supplying 12V CAT 6 Flashers to Force Motor for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V CAT 6 flashers assembly after **Low Temperature Operating test**.

TEST SPECIFICATION:

Place the Electronic Component in the constant temperature chamber under room temperature. Connect the power supply device, input and

Loading devices outside the chamber so the test specimen can be operated. Lower the temperature gradually when the temperature reaches

-20°C. (As per Drg.) Keep the Electronic Component for 1 ± 0.5 hours under the temperature. Then operate the Electronic Component for 70 ± 2 hours under the temperature. Take the Electronic Component out of the chamber remove the water drops from the surface and keep it for 2

Hours at room temperature. Then conduct the performance and functional check of the Electronic Component..

ACCEPTANCE CRITERIA:

All function of the flasher performs as designed before and after the test.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:09	O	O	O
S:10	O	O	O

O: OK

X: NG

Δ: To be confirmed

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements

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Prepared By	Approved By



PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	10-16 V
2	Test voltage	13.5 V
3	Flash rate(Turn signal/Hazard warning)	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST PROCEDURE:

The test performed with all electrical connections with a lamp load of 3 x 21 W Lamp + 1x 2W LED and a voltage of 13.5 V is applied for the duration of 72 Hrs.

TEST SETUP:



TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL

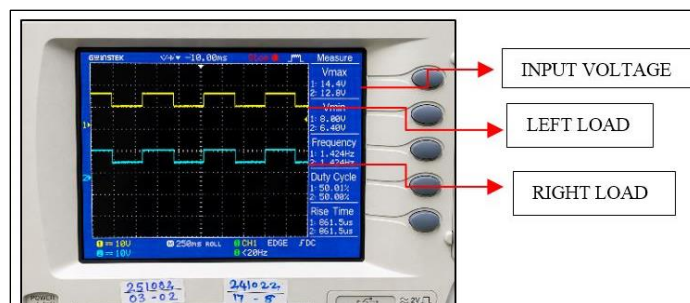


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WAVEFORM DATA:



TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:09,10	
Turn Signal RH	3 x 21 W Lamp + 1x 2W LED	85	OK
Turn Signal LH	3 x 21 W Lamp + 1x 2W LED	85	OK
Hazard Mode	6 x 21 W Lamp + 2x 2 W LED+	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED	171	OK
Duty cycles for RH&LH	3 x 21 W Lamp + 1x 2 W LED	49%	OK
Duty cycles for Hazard Mode	3 x 21 W Lamp + 1x 2W LED	49%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:09,10	AFTER TEST SAMPLE – 09,10

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.